

Pore-scale simulation of mixing-induced calcium carbonate precipitation and dissolution in a microfluidic pore network

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[1] We develop a 2-D pore scale model of coupled fluid flow, reactive transport, and calcium carbonate (CaCO_3) precipitation and dissolution. The model is used to simulate transient experimental results of CaCO_3 precipitation and dissolution under supersaturated conditions in a microfluidic pore network (i.e., micromodel) in order to improve understanding of coupled reactive transport systems perturbed by geological CO_2 injection. In the micromodel, precipitation is induced by transverse mixing along the centerline in pore bodies. The reactive transport model includes the impact of pH upon carbonate speciation and a CaCO_3 reaction rate constant, the effect of changing reactive surface area upon the reaction, and the impact of pore blockage from CaCO_3 precipitation on diffusion and flow. Overall, the pore scale model qualitatively captured the precipitate morphology, precipitation rate, and maximum precipitation area using parameter values from the literature. In particular, we found that proper estimation of the effective diffusion coefficient (D_{eff}) and the reactive surface area is necessary to adequately simulate precipitation and dissolution rates. In order to match the initial phase of fast precipitation, it was necessary to consider the top and bottom of the micromodel as additional reactive surfaces. In order to match a later phase when dissolution occurred, it was necessary to increase the dissolution rate compared to the precipitation rate, but the simulated precipitate area was still higher than the experimental results after ~ 30 min, highlighting the need for future study. The model presented here allows us to simulate and mechanistically evaluate precipitation and dissolution of CaCO_3 observed in a micromodel pore network. This study leads to improved understanding of the fundamental physicochemical processes of CaCO_3 precipitation and dissolution under far-from-equilibrium conditions.

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1. Introduction

[2] The formation of calcium carbonate plays an important role in many natural and engineered systems, including geological CO_2 storage [Johnson *et al.*, 2004; Rochelle *et al.*, 2004], biogeochemical behavior in marine environments [Riebesell *et al.*, 2000], biomineralization of products from microorganisms [De Yoreo and Vekilov, 2003], growth of calcite concretions in fractured and porous media [Mozley and Davis, 2005], and the formation of scale in boilers and cooling towers [Richter *et al.*, 1989]. Knowledge of the kinetics and mechanisms of crystal growth facilitates prediction of mineralization associated with these processes, and can guide the design of engineered systems. Within hydrogeochemical, engineered, or biological systems, mineralization processes occur at different spatial and temporal scales. In particular, geochemical

reactions in porous media occur at the molecular scale, and mineral products affect flow and mixing processes at the pore and continuum scale. These have consequences for predicting reactive transport processes in porous media, such as those during geological storage of CO_2 . Mineral precipitation and dissolution has been extensively studied in well-mixed batch systems; however, there have been few investigations for transport conditions where mass transfer and dispersion limits the mixing of reactants. As a result, a mechanistic understanding of mineral formation and dissolution rates in porous media is lacking, thus limiting the accuracy of quantitative assessments required for scale-up to field systems.

[3] There is extensive literature that details indirect measurement approaches (e.g., measuring solution chemistry, weight, and/or size) to quantify calcium carbonate (CaCO_3) crystallization in well-mixed systems under a wide range of chemical compositions and conditions [Plummer *et al.*, 1978; Inskeep and Bloom, 1985; Blum and Lasaga, 1987; Hostomsky and Jones, 1991; Shiraki and Brantley, 1995]. These studies have been used to develop quantitative representations of calcite growth rate laws commonly used in reactive transport models for porous and fractured media. However, these models have not been confirmed through

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observations of the actual growth processes at mineral surfaces. Recently, in situ measurement techniques have been used to elucidate crystal growth mechanisms from highly supersaturated solutions in well-mixed systems [Pontoni *et al.*, 2003; Rieger *et al.*, 2007; Liu *et al.*, 2010; Kellermeier *et al.*, 2010]. In particular, these works investigated the role of nano-size amorphous calcium carbonate (ACC) in the growth of calcium carbonate crystals. ACC is the thermodynamically least stable among all calcium carbonate (CaCO_3) polymorphs, followed by vaterite, aragonite, and calcite. Under highly supersaturated conditions, ACC nanoparticles with a size of ~ 30 nm are formed within a minute; subsequently, these particles grow and transform with time to more stable crystalline forms such as vaterite crystals [Pouget *et al.*, 2009; Liu *et al.*, 2010], and eventually calcite [Ogino *et al.*, 1987; Rieger *et al.*, 2007].

[4] Significant efforts have also been made to elucidate crystal growth mechanisms from supersaturated solutions under representative hydrodynamic conditions using in situ measurement techniques. Mineral growth and dissolution were measured using in situ atomic force microscopy (AFM) [Teng *et al.*, 2000; De Yereo *et al.*, 2009; Stack and Grantham, 2010] and X-ray computed microtomography [Gouze *et al.*, 2003; Noiriel *et al.*, 2004], and mineral morphology and type were measured using a combination of reflected differential interference contrast (DIC) microscopy and backscattering Raman-spectroscopy [Zhang *et al.*, 2010]. Stack and Grantham [2010] determined using AFM that growth rates of calcite and calcite morphology are strongly affected by the ratio of calcium to carbonate concentration, implying that calcite growth patterns will be affected in a spatially varying concentration environment such as porous media. These findings are supported by results from Zhang *et al.* [2010], where Raman spectroscopy was used to show that more vaterite relative to calcite was formed at higher calcium and carbonate concentrations in a microfluidic pore network, and that vaterite crystals were smaller than calcite crystals. These novel experimental and theoretical results under hydrodynamic conditions have the potential to enhance our ability to simulate reactive transport in the subsurface.

[5] A number of pore scale models for flow and reactive transport relevant to geological storage of CO_2 have been developed using lattice Boltzmann (LB) [Kang *et al.*, 2003, 2006, 2010], smooth particle hydrodynamics (SPH) [Tartakovsky *et al.*, 2007, 2008, 2009], direct numerical simulation [Flukiger and Bernard, 2009], and pore network models [Li *et al.*, 2006; Kim *et al.*, 2011]. Both LB and SPH are suitable for complex geometries and a variety of reactive surface boundaries, and have been applied to fluid flow and multicomponent reactive transport problems. These modeling studies characterized qualitative features of reactive transport in a simple numerical domain over a wide range of dimensionless Peclet (Pe) and Damköhler (Da) numbers [Kang *et al.*, 2006, 2007; Tartakovsky *et al.*, 2007, 2008]. In particular, the SPH model was used to characterize mixing-induced precipitation in a meter-long tank experiment over a range of the Peclet and Damköhler numbers where the characteristic length scale of precipitation was on the order of the average sand grain diameter along the mixing line [Tartakovsky *et al.*, 2008]. However, the SPH model did not match changes in pH, carbonate species

concentrations, or the saturation index value in the tank experiment. Pore network reactive flow models [Li *et al.*, 2006; Kim *et al.*, 2011] remain a practical approach for simulation since they are flexible, computationally efficient, and can incorporate reaction rate laws and crystal growth patterns into the modeling framework. However, these attempts are based on simplified assumptions for crystal growth at the fluid-reaction interface, and do not fully account for pore scale processes under hydrodynamic conditions occurring in porous media.

[6] Recent developments of in situ measurement techniques for pore scale-reactive transport experiments provide a unique opportunity to test and validate pore scale-modeling approaches. For example, recent attempts to simulate CaCO_3 precipitation and dissolution in porous media [Emmanuel and Berkowitz, 2005; Tartakovsky *et al.*, 2008; Kang *et al.*, 2010] revealed the importance of porous media structure and a heterogeneous reaction rate constant. In porous and fractured media, the numerical studies suggest that diffusion-controlled precipitate can armor fractures, and limit the amount of mixing and subsequent reaction [Steefel and Lichtner, 1994, 1998; Kang *et al.*, 2010]. However, the theory that captures these effects has not been quantitatively compared against experimental data. Also, several key processes have not been evaluated by pore scale models, including the effect of mineral formation on diffusion [Navarre-Sitchler *et al.*, 2009] and reactive surface area [Navarre-Sitchler and Brantley, 2007]. Pore scale or micro-scale models are typically used to gain fundamental insights into physical behavior so that more accurate and robust upscaled constitutive models can be developed for the continuum scale [Whitaker, 1999; Wood, 2009; Kim *et al.*, 2011]. The development of pore scale models also provides the opportunity to more accurately simulate results at practical length scales in a hybrid approach in which pore- and continuum-scale equations are combined using either a domain-decomposition approach [Balhoff *et al.*, 2008; Battiato *et al.*, 2011] or dimension-reduction methods based on numerical closure with an explicit averaging of information from pore scale simulations [Tartakovsky and Scheibe, 2011].

[7] The objectives of this study are to present a new pore scale-reactive transport model for calcium carbonate precipitation and dissolution, to evaluate it with published experimental results from a microfluidic pore network experiment, and to identify mechanisms that are poorly captured with existing theory. The new pore scale model consists of three coupled modules: a lattice Boltzmann (LB) module for Stokes flow in the pore network, a finite volume multicomponent reactive transport module for the advection and diffusion of dissolved species, and a new pore scale crystal growth module that determines the amount of precipitate in the pore space. Model results are compared to experimental results by Zhang *et al.* [2010], who measured calcium carbonate precipitation and dissolution in a two-dimensional pore network (i.e., micromodel) along a transverse mixing zone between influent solutions of calcium chloride and sodium carbonate. We evaluate how the effective diffusion coefficient, reaction rate, and reaction surface area affect agreement between model simulations and experimental results. The implications of the pore scale modeling results are discussed.

2. Micromodel Experiment

[8] Homogeneous micromodels with a uniform distribution of cylinders were used to perform transverse-mixing induced CaCO_3 precipitation and dissolution experiments [Zhang *et al.*, 2010; Dehoff, 2010]. A detailed description of the fabrication and assembly process is given by Chomsurin and Werth [2003] and Willingham *et al.* [2008], and details of experimental results are given by Zhang *et al.* [2010]. The micromodel pore network is $1 \text{ cm} \times 2 \text{ cm}$ (width $\times$ length), consisting of cylindrical posts $300 \mu\text{m}$ in diameter, $180 \mu\text{m}$ pore bodies, and $40 \mu\text{m}$ pore throats. The depth of a micromodel pore is $20 \mu\text{m}$. Porosity is ~ 0.39 and water was continuously injected through two separate inlets, resulting in a constant Darcy velocity of 1.25 cm min^{-1} .

[9] The experiments consisted of injecting solutions containing equimolar concentrations of CaCl_2 and Na_2CO_3 through two separate inlets, transverse diffusion caused the two solutions to mix along the center of the micromodel, resulting in CaCO_3 precipitation. In this work, we focus on modeling CaCO_3 precipitation and dissolution in the first three pore bodies closest to the inlets of the pore network. We evaluated transient CaCO_3 precipitation and dissolution with 25 mM influent concentrations of total calcium and carbonate until the amount of precipitate reached quasi-steady state (i.e., the precipitate area was constant over time).

[10] An experimental image of precipitates at steady state for the 25 mM inlet concentrations is shown in Figure 1a. Precipitation occurred mostly along the centerline within one pore space transverse to the primary flow direction, with some large crystals off the centerline. The transverse width of the precipitate line tends to an increase with distance from the inlet. Two observations relevant to this work are that precipitation occurred upstream prior to downstream, and the precipitate is slightly asymmetric in the transverse direction downstream as shown in the first

pore body. Using backscattering Raman spectroscopy, Zhang *et al.* [2010] identified that larger crystals are calcite, and clusters of smaller particles along the centerline are vaterite. In this study, only vaterite formation is considered since it is the dominant precipitate phase under the experimental conditions considered.

[11] The experimental images used in this study were taken using reflected differential interference contrast microscopy [Zhang *et al.*, 2010]. The image intensity of CaCO_3 precipitates (shown in black or dark gray) is lower than that of the cylinder and pore spaces, but a shadow around the cylinder has a similar intensity (Figure 1a). In this study, new masks were created to cover the cylindrical posts and their shadows over the image region considered (i.e., the first three pore spaces), and then a threshold value was determined for each image to calculate the precipitate areas (i.e., the number of pixels). This method results in a 2-D precipitate area. The measured precipitate areas in the first pore body and entire numerical domain (i.e., the first three pore spaces) are shown in Figure 1b. An important result for this study is that the precipitate area increases rapidly due to fast precipitation, and then decreases due to fast dissolution within 13 min until a relatively constant but lower plateau is reached over 3–4 h. We denote the rapid increase in precipitate area followed by a decrease *overshoot*. In this study, the characteristic of the precipitate area over time (i.e., overshoot) and patterns of the precipitate distribution in the 2-D micromodel are investigated.

[12] A basic assumption of this work is that many of the important features in the micromodel experiments can be captured with a 2-D numerical model, because all experimental images were measured in 2-D, and laser confocal microscopy images showed that mineral precipitates span the entire depth of micromodel pores at the end of the experiment [Zhang *et al.*, 2010]. In addition, because of the

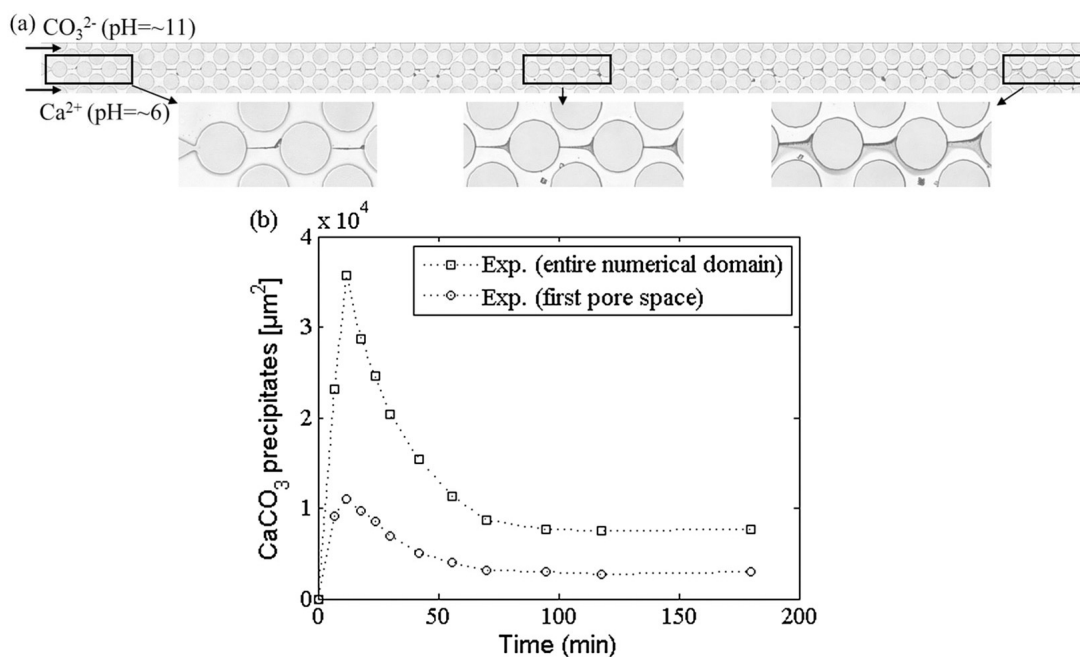


Figure 1. (a) Image of CaCO_3 precipitates in the center of the micromodel at 118 min for 25 mM influent concentrations and (b) CaCO_3 precipitate area over time.

shallow depth (20 μm) of the micromodel, the diffusion time-scale for solute concentration to be uniform with depth is relatively small (~ 0.1 s) compared to the fluid residence time (~ 2 min). Hence, the reaction is dominantly affected by horizontal transverse-mixing of two reactants via diffusion along the centerline, and this work allows us to isolate and quantify the effects of diffusion and reaction parameters on the heterogeneous reaction under flowing conditions ($\text{Pe} = \sim 100$).

3. Model Description

[13] The model consists of three components: a flow module, a multicomponent reactive transport module, and a precipitation and dissolution module. We assume that the flow field reaches steady state instantaneously at each time step because the residence time in the numerical domain is short. We also assume that the geometry of the solid phase is constant during each time step when reactive transport occurs, until the reaction alters the liquid phase (or solid phase) in any grid cell.

3.1. Flow Field

[14] The lattice-Boltzmann method (LBM) is used to solve the steady state Stokes equation for interstitial pore velocities as described in our previous work [Knutson *et al.*, 2005]. A standard single-relaxation time model (or Bhatnagar-Gross-Krook [BGK]) [e.g., Hou *et al.*, 1995; Noble *et al.*, 1995] was used. No-slip boundary conditions are imposed at all solid surfaces (i.e., original cylinder posts in the micromodel and precipitated minerals) via a bounce-back boundary condition. Along the horizontal boundaries in the 2-D domain, no-slip boundary is imposed. Fixed pressure boundary conditions are imposed along the inlet ($x = 0$) and outlet ($x = L$) and the resulting steady flow field is scaled to the Darcy velocity used in the experiment. A new steady flow field is solved each time the pore structure changes due to precipitation and dissolution.

3.2. Reactive Transport

[15] We solve the steady state advection-diffusion reaction equation using the velocity field from the LBM. We assume solute concentrations do not change over the time interval between updates of the volumetric fraction of precipitates. We assume that homogeneous reactions are instantaneous, and that all heterogeneous reactions at the mineral surface are kinetically controlled [Lichtner, 1996; Steefel *et al.*, 2005; Kang *et al.*, 2006]. The heterogeneous reaction is treated through the boundary condition on the reactive surface (Γ_s). The reactive transport equations in terms of the total concentration of the j -th primary species (Ψ_j) in the liquid phase (Ω_p) and heterogeneous reaction on the reactive surface (Γ_s) are given by

$$\nabla \cdot (\mathbf{u}\Psi_j - D\nabla\Psi_j) = 0 \quad \text{in } \Omega_p, \quad (1)$$

$$D \frac{\partial \Psi_j}{\partial \mathbf{n}} = -I_m \quad \text{on } \Gamma_s, \quad (2)$$

$$\Psi_j = C_j + \sum_{i=1}^{N_{eq}} \nu_{ji} C_i, \quad (3)$$

where $\mathbf{u}$ is the steady velocity field from LBM, D is the diffusion coefficient, ν_{ji} are the stoichiometric coefficients, $\mathbf{n}$ is the outer unit normal to the interface, I_m is the precipitation-dissolution rate, C_j and C_i are solute concentrations of primary and secondary species in the liquid phase, respectively, and N_{eq} is the number of secondary species (subject to equilibrium reaction). In this study, the same diffusion coefficient for all species is used to avoid the violation of charge balance [Lichtner, 1985; Lichtner and Kang, 2007; Li *et al.*, 2007]. A finite volume method (FVM) with a second-order approximation for advection and diffusion is used to numerically solve the system, and the resulting equations with the heterogeneous reaction (equation (2)) are linearized by the enhanced sequential iterative approach (SIA-1) [Tebe-Stevens *et al.*, 1998]. Fixed concentrations and a zero gradient boundary conditions were used at the inlet and outlet, respectively.

3.3. Reaction Kinetics

[16] Three primary species, Ca^{2+} , H^+ , and CO_3^{2-} , and three secondary species, OH^- , HCO_3^- , and $\text{H}_2\text{CO}_{3(\text{aq})}$ (where $\text{H}_2\text{CO}_{3(\text{aq})}$ represents the sum of H_2CO_3 and $\text{CO}_{2(\text{aq})}$) are considered to model CaCO_3 precipitation and dissolution. As mentioned previously, only the formation of vaterite is considered. The secondary species concentrations (C_i) are given through the mass action law as functions of the primary species concentrations:

$$C_i = \gamma_i^{-1} K_i \prod_{j=1}^{N_c} (\gamma_j C_j)^{\nu_{ji}}, \quad (4)$$

where γ_i is the activity coefficient and K_i is the equilibrium constant. The activity coefficient is computed from the extended Debye-Hückel (D-H) equation:

$$\log \gamma_i = -\frac{Az_i^2 \sqrt{I}}{1 + aB\sqrt{I}}, \quad (5)$$

where A and B are constant for water at 25°C , a is the ion size parameter, z is the valence, and I is the ionic strength. All constant values are taken from Helgeson and Kirkham [1974]. The ionic strength is defined as

$$I = \frac{1}{2} \sum_i m_i z_i^2, \quad (6)$$

where m_i is the molality (where $C_i = \rho m_i$). The nonreactive species (Na^+ and Cl^-) concentrations are solved independently to compute activity coefficients. The system is fully saturated with water (i.e., closed to the atmosphere), and the density of the aqueous phase is equal to pure water (i.e., dilute solution).

[17] The CaCO_3 precipitation-dissolution rate I_m has the form [Chou *et al.*, 1989]:

$$I_m = -k_{cc} (1 - \Omega) = -(k_1 a_{\text{H}^+} + k_2 a_{\text{H}_2\text{CO}_3} + k_3) \left(1 - \frac{Q_{cc}}{K_{sp}}\right), \quad (7)$$

where k_{cc} is the overall reaction rate constant, Ω is the supersaturation index with respect to CaCO_3 , which is defined as the ratio of ion activity product (Q_{cc}) to the saturation product (K_{sp}) of CaCO_3 , k_1 , k_2 , and k_3 are the reaction rate constants, a_i is the activity of species i , and Q_{cc} ($= a_{\text{Ca}^{2+}} a_{\text{CO}_3^{2-}}$) is the ion activity product (IAP) of calcium and carbonate ions. The values of k_1 , k_2 , and k_3 are 8.9×10^{-1} , 5.01×10^{-4} , and $6.6 \times 10^{-7} \text{ mol m}^{-2} \text{ s}^{-1}$, respectively [Chou *et al.*, 1989]. The overall reaction rate constant observed in the work of Chou *et al.* [1989] is relatively large, compared to other values reported in the literature [e.g., Shiraki and Brantley, 1995]. This value is used to account for a highly supersaturated condition in the micromodel experiment, since the growth of calcium carbonate crystals with AFM by Teng *et al.* [2000] was observed to increase with supersaturation index. It is assumed that CaCO_3 precipitation and dissolution occur only at reactive surfaces (equation (2)). The overall reaction rate (R_s) is the product of I_m and the reactive surface area in the grid cell. All equilibrium constants and vaterite properties are presented in Table 1.

3.4. Update of Solid Phase

[18] The solid precipitate phase is assumed immobile, and the volume change of the precipitate is given by

$$\frac{\partial V_m}{\partial t} = \bar{V}_m s_{cc} I_m, \quad (8)$$

where V_m and $\bar{V}_m$ are the volumetric fraction and molar volume of CaCO_3 , and s_{cc} is the specific surface area of the reactive surface for CaCO_3 . Similar approaches have been previously used for multicomponent reactive transport systems [Kang *et al.*, 2006, 2010]. The volumetric fraction of CaCO_3 after each time step is updated explicitly by

$$V_m(t + dt) = V_m(t) + \bar{V}_m s_{cc} I_m \cdot dt. \quad (9)$$

[19] The value of the time interval (dt) is determined as a fraction of the maximum growth or dissolution rate in the system. If the calculated time interval is greater than a

maximum value specified, depending upon the experimental growth rate, the maximum value is used instead in order to avoid a rapid change of the solid phase within one time step. The maximum value of 0.1 min was used at the beginning of the simulation, with a gradual increase to 2 min.

[20] Although the micromodel is considered a 2-D system, precipitate can form on the horizontal top and bottom surfaces as well as on the vertically oriented surfaces of the cylindrical posts. To account for this, each numerical grid cell was conceptualized to have a depth equal to that of the micromodel (i.e., $dz = 20 \text{ } \mu\text{m}$), and the reactive surface area was calculated by assuming CaCO_3 could precipitate both on the bottom and top of the grid cell (i.e., to simulate precipitation on the top and bottom of the micromodel) (as shown in Figure 2) and on vertical walls for grid cells adjacent to a cylindrical post. Mineral precipitates block the pore space, so we use a threshold value for V_m ($V_{m,TV}$), above which it is assumed that there is no flow through a grid cell. Our preliminary numerical simulation results showed that the simulation results with $V_{m,TV}$ below 0.5 or above 0.75 do not agree with experimental results. Within the range from 0.5 to 0.75, the results were only mildly sensitive to $V_{m,TV}$; therefore, $V_{m,TV}$ was set equal to 0.6. The flow field is updated if V_m becomes greater than $V_{m,TV}$ in any grid cell due to mineral reactions after updating the volume of CaCO_3 .

[21] When V_m reaches 1 in a grid cell, the vertical face of the grid cell ($dz \times dx$ or $dz \times dy$) is the reactive surface for precipitation if $\Omega > 1$ in an adjacent cell, or for dissolution if $\Omega < 1$ in an adjacent cell without CaCO_3 (i.e., $V_m = 0$) until V_m is less than $V_{m,TV}$. To account for differences in precipitation and dissolution rates, a dissolution factor (d_f) is introduced as a multiplier in the overall reaction rate; a dissolution factor of 1 means precipitation and dissolution rates are equal. The overall reaction rate (R_s) in a grid cell and volume change of the precipitate are described by

$$R_s = s_{cc} d_f I_m, \quad (10)$$

$$\frac{\partial V_m}{\partial t} = \bar{V}_m s_{cc} d_f I_m, \quad (11)$$

$$V_m(t + dt) = V_m(t) + \bar{V}_m s_{cc} d_f I_m \cdot dt. \quad (12)$$

[22] In the micromodel, d_f is applied after the precipitate along the centerline blocks the pore space, so the region with $V_m < 1$ has $\Omega < 1$. Since the overall reaction rate (equation (10)) is a product of several terms, a value of d_f greater than 1 can correspond to either an increase in the dissolution rate constant or an increase in the reactive surface area. In this work, we are unable to distinguish these two mechanisms, so they are evaluated as a lumped parameter.

[23] Diffusion is reduced in a grid cell occupied by precipitates due to an increase in tortuosity (τ). Hence, we incorporate the effect of τ as V_m changes due to mineral precipitation and dissolution by adapting a form of Archie's law [Archie, 1942]. Archie's law can be expressed as

$$D_{\text{eff}} = D \cdot \tau = D \cdot (1 - V_m)^n, \quad (13)$$

Table 1. Parameter Values Used in Simulations

	Value	SI Units
Diffusion coefficient (D) ^a	9.0×10^{-10}	($\text{m}^2 \text{ s}^{-1}$)
Exponent n	2	—
Reaction rate constant k_1	8.9×10^{-1}	($\text{mol m}^{-2} \text{ s}^{-1}$)
Reaction rate constant k_2	5.01×10^{-4}	($\text{mol m}^{-2} \text{ s}^{-1}$)
Reaction rate constant k_3	6.6×10^{-7}	($\text{mol m}^{-2} \text{ s}^{-1}$)
Threshold for flow, $V_{m,TV}$	0.6	—
Solubility product, K_{sp}	1.832×10^{-2}	($\text{mol}^2 \text{ m}^{-6}$)
Molar volume, $\bar{V}_m$	3.747×10^{-5}	($\text{m}^3 \text{ mol}^{-1}$)

^aFrom Li and Gregory [1974] and Poisson and Papaud [1983].

Reaction rate constants are from Chou *et al.* [1989].

Parameter values are default values unless separately stated.

The total concentrations for each primary species are

$$\Psi_{\text{H}^+} = C_{\text{H}^+} - C_{\text{OH}^-} + C_{\text{HCO}_3^-} + 2C_{\text{H}_2\text{CO}_3}$$

$$\Psi_{\text{CO}_3^{2-}} = C_{\text{CO}_3^{2-}} + C_{\text{HCO}_3^-} + C_{\text{H}_2\text{CO}_3}$$

$$\Psi_j = C_j \text{ for } j = \text{Ca}^{2+}, \text{Na}^+, \text{Cl}^-.$$

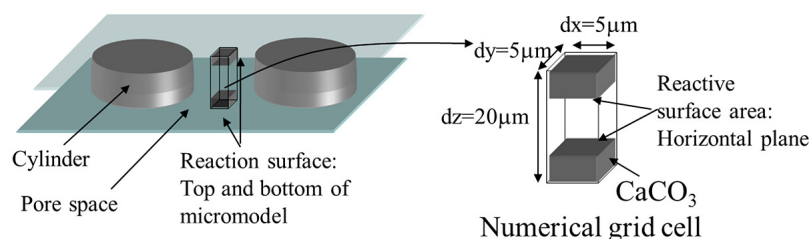


Figure 2. Schematic of reaction model. A numerical grid cell has a dimension of $5 \times 5 \times 20 \mu\text{m}^3$. The horizontal grid area is initially used as the reactive surface area and the reactive surface area evolves as a function of the volumetric content of calcium carbonate as shown in gray blocks (right).

where n represents the cementation exponent in natural materials [Navarre-Sitchler *et al.*, 2009]. In this study, the exponent n was varied from 0 to 3 in order to test the effect of the diffusion coefficient on reaction at the pore scale. In natural materials, the value for n was observed to range from 2 to 2.2 [Boving and Grathwohl, 2001; Navarre-Sitchler *et al.*, 2009].

3.5. Model Application Scenarios

[24] A uniform grid spacing of $5 \mu\text{m}$ is used for both the LB and FV methods, which results in 60 grid cells across a grain and seven grid cells across a pore throat. Additional simulation results of precipitate in the first pore space with a $1 \mu\text{m}$ grid were not significantly different (i.e., $<3\%$ in precipitate mass) from the results with a $5 \mu\text{m}$ grid; hence, only the results with a $5 \mu\text{m}$ grid are presented in this study. As mentioned, the depth of the numerical domain (i.e., one grid in the vertical direction) is $20 \mu\text{m}$. The grid spacing of $5 \mu\text{m}$ has previously been used to simulate different biochemical reactions in an identical pore structure [Knutson *et al.*, 2005; Willingham *et al.*, 2008, 2010].

[25] Representative chemical and physical properties used in the simulations are listed in Table 1. Inlet concentrations for total concentrations of calcium, carbonate, and hydrogen ions are computed based upon the experimental values of the inlet pH and concentrations of CaCl_2 and Na_2CO_3 , equilibrium constants, and activity coefficients computed using equation (5). The total carbonate concentration in CaCl_2 solution was assumed negligible ($\ll 10^{-5} \text{ mol L}^{-1}$) compared to that in a Na_2CO_3 solution because the former solution was degassed in a vacuum. As mentioned earlier, the diffusion coefficient of all aqueous species is the same and set to $0.9 \times 10^{-5} \text{ cm}^2 \text{ s}^{-1}$ [Li and Gregory, 1974; Poisson and Papaud, 1983].

[26] Five separate cases with different parameter sets were used to test the impact of the following key parameters: effective diffusion coefficient, overall reaction rate, and dissolution factor. There are summarized in Table 2. Case 1 is considered the base case. For this case, the exponent n for D_{eff} was set equal to 2, reaction kinetic constants are given by Chou *et al.* [1989], and the dissolution factor was set to 1. The exponent n was changed to 0 and 3 for cases 2 and 3, respectively. When n is equal to zero the diffusion coefficient is constant and not affected by pore blockage due to precipitation. Case 4 has different k_1 , k_2 , and k_3 values, each equal to twice the base case values in case 1. For case 5, the dissolution factor is changed to better match the data.

4. Results

4.1. Base Case (Case 1)

[27] Images of CaCO_3 precipitates in the micromodel and corresponding simulation results at 13, 18, and 118 min are shown in Figure 3. The measured and simulated precipitate areas in the entire numerical domain (i.e., the first three pore spaces) are also shown in Figure 4a. The simulated precipitate area was determined for grid cells with a value of V_m greater than 0.05. Other threshold values were evaluated, but resulted in a poorer fit of the experimental data using base case parameter values. Transverse mixing between calcium and carbonate ions created a supersaturated zone along the central mixing line, yielding simulated Ω values in the first pore body of ~ 2100 before a reaction occurred and up to ~ 800 after a reaction occurred (Figure 4b). This resulted in rapid precipitation in the first 13 min. In each pore body, more diffusive mixing occurred with distance, resulting in a wider precipitate zone downstream than upstream. Experimental images at 13 and 18 min show that the precipitation area in the first pore space was quite similar, but the darkness of the precipitate zone increased, suggesting that V_m increased over time along the centerline. The precipitation patterns in the other pore bodies followed the pattern in the first pore body. Precipitation was followed by dissolution over ~ 40 min (Figures 3c and 4a). Eventually, the precipitate area reached an asymptotic level with a very thin precipitate line along the centerline of each pore body (Figure 3c), which lasted over 3 h until the end of the experiment. Experimental images suggest that as V_m increases, diffusive mixing of Ca^{2+} and CO_3^{2-} was reduced, and this led to dissolution along the outer edges of the precipitate zone.

[28] Simulation results in Figures 3 and 4 appear to capture both the initial rate of precipitation and the precipitation

Table 2. Conditions Used in Simulation Cases

Case	Exponent Constant, n	Kinetic Rate ^a ($\text{mol cm}^{-2} \text{ s}^{-1}$)	Dissolution Factor
1 ^b	2	k_{cc}	1
2	0	k_{cc}	1
3	3	k_{cc}	1
4	2	$k_{cc} \times 2$	1
5	2	k_{cc}	300

^aThe default values are from Chou *et al.* [1989]. k_{cc} is the overall reaction rate constant in equation (7).

^bCase 1 is the base case. Other cases use conditions defined in case 1 except where conditions are described.

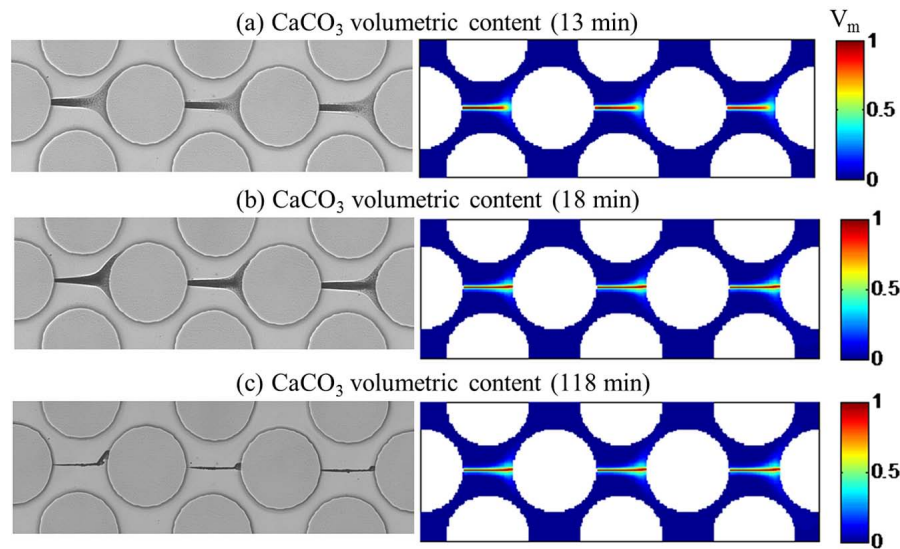


Figure 3. Image of CaCO_3 precipitates and simulated CaCO_3 volumetric content at (a) 13 min, (b) 18 min, and (c) 118 min, for case 1.

morphology observed in the micromodel. Before pore blocking, the pH increased dramatically in the mixing zone due to reaction (i.e., carbonate ion depletion) and diffusion (Figure 4c). This indicates that the effect of pH on the precipitation rate is likely to be insignificant as long as diffusion is allowed across the centerline. However, simulations were not able to capture dissolution after ~ 15 min. After 24 min, precipitates were sufficiently dense across each pore body, such that Ω became < 1 off the centerline (i.e., undersaturated conditions). However, simulated dissolution is much slower compared to the experimental results (Figure 4a), indicating that less diffusive mixing after pore blockage, greater kinetic rate constants, and/or a higher reactive surface area during dissolution (equations (10)–(12)) are required.

4.2. Effective Diffusion Coefficient (Cases 1–3)

[29] The influence of the effective diffusion coefficient, i.e., the parameter n (see equation (13)), on the amount of precipitate over time, and the corresponding experimental

image after 13 min are shown in Figure 5. The measured and simulated precipitate areas are also shown (Figure 5f). For the default case of $n = 2$ (case 1), D_{eff} decreases as V_m increases and this limits the amount of precipitate and yields good agreement with the experimental data up to ~ 13 min. As the n -value decreases, D_{eff} decreases less as V_m increases, resulting in more precipitation (i.e., a higher-precipitation rate) compared to the experimental results. As the n -value increases, D_{eff} decreases more as V_m increases, resulting in less precipitation due to a smaller reaction zone. However, the range of n -values evaluated does not adequately capture the decrease in the precipitate area that occurs after ~ 13 min.

4.3. Overall Reaction Constant (Cases 1 and 4)

[30] The influence of overall reaction constant (k_{cc}) on precipitation and dissolution and a corresponding experimental image after 13 min are shown in Figures 5d and 5e. The measured and simulated precipitate areas are also shown (Figure 5f). The k_{cc} value for case 4 is multiplied

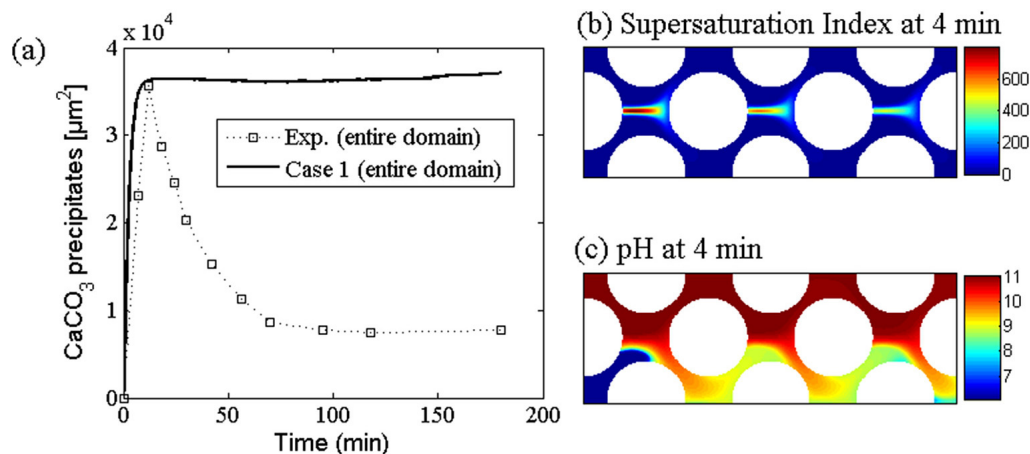


Figure 4. (a) Comparison of CaCO_3 precipitates area in the entire numerical domain, (b) supersaturation index at 4 min, and (c) pH at 4 min (case 1).

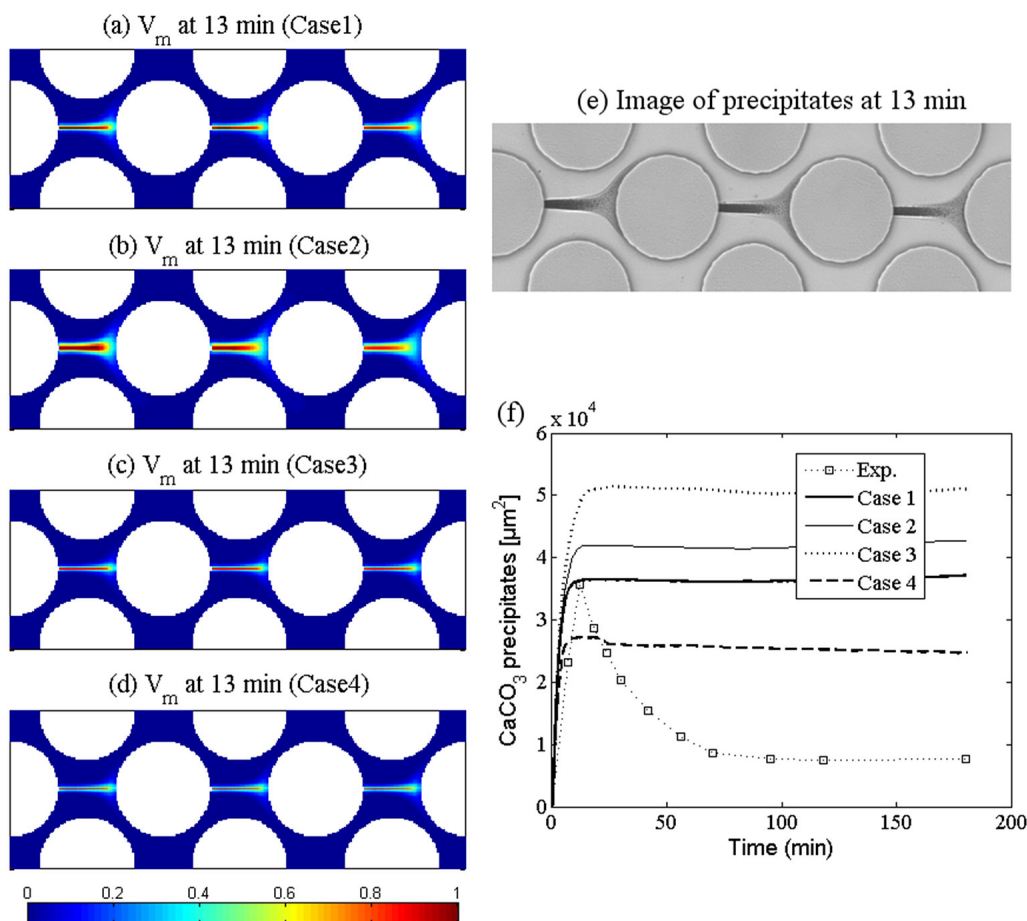


Figure 5. Effect of the exponent n and overall reaction constant on CaCO_3 volumetric content (V_m). (a) $n = 2$ (case 1), (b) $n = 0$ (case 2), (c) $n = 3$ (case 3), and (d) $k_{cc} \times 2$ (case 4). (e) Experimental image of precipitates at 13 min, and (f) measured and simulated precipitate areas over time.

by a factor of two, compared to that for case 1. As k_{cc} increases, precipitation occurs faster, particularly along the centerline, resulting in a reduction of diffusion (i.e., mixing). Hence, increasing k_{cc} results in decreasing the maximum precipitate area. After the maximum precipitation area is reached, some dissolution is predicted but at a much slower rate than the experimental data. Hence, the reaction rate from Chou *et al.* [1989] appears to adequately capture the growth rate at early time, as well as the maximum area of precipitation for the calibrated threshold value of $V_m = 0.05$, but the extent of dissolution is not adequately captured.

4.4. Dissolution Factor (Case 5)

[31] Dissolution factors (d_f) ranging from 10 to 1000 were evaluated and the results for d_f greater than 300 were nearly identical. The simulation results with d_f equal to 300 best matched the experimental data, and are presented with experimental images at three different times in Figure 6. The measured and simulated precipitate areas are shown in Figure 7a. In contrast to case 1, the thickness and area of precipitate along the centerline after 13 min for case 5 decreases more quickly, and the precipitate area matches the experimental data until 30 min. In addition, as in the

experimental data, the model predicts that dissolution below the centerline occurs more quickly than dissolution above the centerline. This is because of the higher pH above the centerline after pore blocking (Figures 7b and 7c), resulting in a higher supersaturation index above the centerline (Figure 7d). Hence, the discrepancy in the precipitate area between the simulation and the experimental results after 30 min is attributed to slow dissolution above the centerline.

5. Summary and Conclusions

[32] A new 2-D pore-scale reactive transport model was used to explore the mechanisms of transverse-mixing induced CaCO_3 precipitation and dissolution in a micromodel experiment [Zhang *et al.*, 2010]. The model considers the effects of pH on carbonate speciation and a CaCO_3 reaction rate constant, of changing reactive surface area on the reaction, and of pore blockage from CaCO_3 precipitation on diffusion and flow. The reactive surface area during precipitation in a grid cell was assumed to include the top and bottom surfaces of the micromodel and the cylinder walls for grid cells adjacent to 2-D grains. In the experiment, the precipitate area increased rapidly due to fast precipitation for ~ 13 min, decreased rapidly due to fast dissolution within 30 min, and then approached a relatively

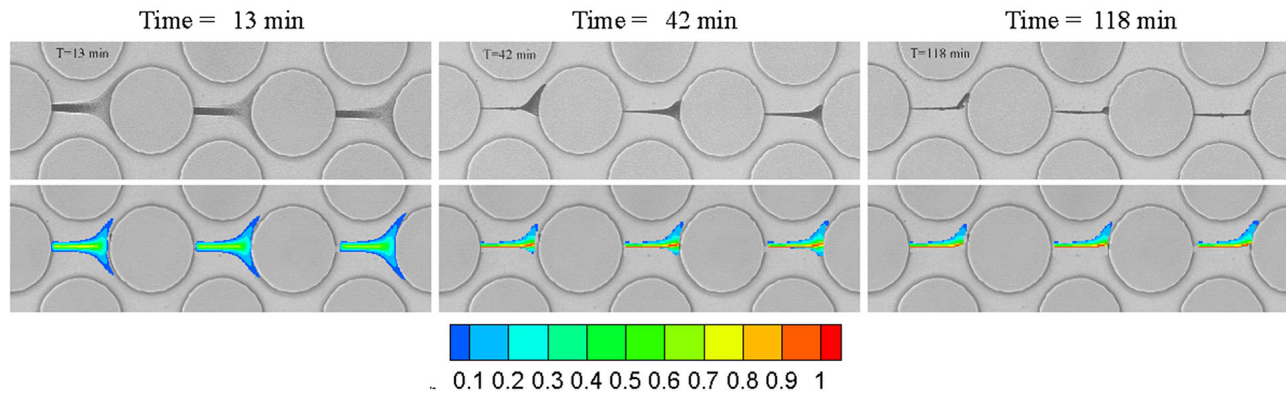


Figure 6. Image of CaCO_3 precipitates (top) and simulated CaCO_3 volumetric content (bottom) at three different times (case 5). A dissolution factor of 300 is used for case 5.

constant but lower plateau over 3–4 h. In simulation results, the grid cells with $V_m > 0.05$ were counted as precipitate. Overall, the pore scale model using parameter values from the literature captured the precipitate morphology, precipitation rate, and maximum precipitation area from the experiment. However, a dissolution factor was required to simulate the rapid decrease in precipitate area after 13 min, and this only allowed the model to capture the precipitate area until 30 min. After this time, the simulated precipitate area leveled off, while the experimental precipitate area continued to decrease.

[33] The effective diffusion parameter (D_{eff}) that accounted for the effect of precipitate on hindered diffusion was expressed as a function of V_m (equation (13)). Greater n -values resulted in slower diffusion with increasing V_m , and less overall precipitate in the pore space. However, values of n between 0 and 3 did not improve the simulation of the rapid dissolution observed in the experiment after

13 min. The value of the overall reaction rate constant, k_{cc} , was increased two times to explore the effects on CaCO_3 precipitation and dissolution. A higher k_{cc} value increased the overall rate of precipitation along the centerline during the first 13 min (equation (7)), and decreased the maximum amount of precipitate due to fast pore blocking, resulting in a poorer fit of the data. However, changing k_{cc} did not improve the simulation of the rapid dissolution observed in the experiment after 13 min.

[34] The need to include a dissolution factor to capture the rapid decrease in the precipitate area after 13 min may be attributed to an increase in the precipitate surface area and the recrystallization process during dissolution. We initially assume the precipitate surface area is equal to the surface area of the micromodel top and bottom and the cylinder walls when a grid cell is adjacent to a 2-D grain. It is somewhat surprising that this simple assumption allows us to capture the precipitation rate during the first 13 min.

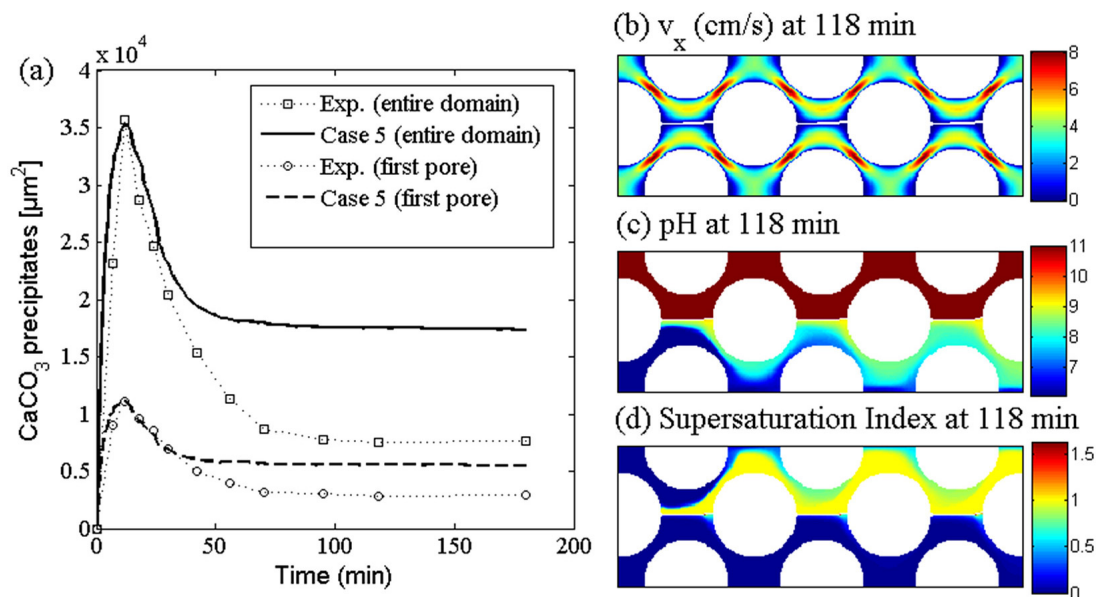


Figure 7. (a) Comparison of CaCO_3 precipitates area in the first pore space (case 5) and the entire numerical domain (case 5), (b) horizontal velocity, (c) pH, and (d) supersaturation index at 118 min (case 5). White color in pore bodies in (b) corresponds to grid cells with $V_m > 0.95$.

However, it is not evident why a different precipitation and dissolution rate are required to match the overshoot until 30 min. Shao *et al.* [2010] observed nanometer-scale pitting on a phlogopite surface during dissolution with supercritical CO₂ using atomic force microscopy. In an earlier reactive flow model, an increase in the reactive surface area over time was considered by using crystal-size distributions that change due to heterogeneous nucleation and the effect of the Ostwald ripening on crystal growth [Steeff and Van Cappellen, 1990]. Rieger *et al.* [2000] used transmission X-ray microscopy to observe that freshly formed vaterite crystals at the micron scale can dissolve completely within 6 min and recrystallize into bigger particles by diffusion. More recently, Pouget *et al.* [2009] used cryotransmission electron microscopy and observed nanoparticles with a diameter of ~30 nm adhere to the surface of vaterite crystals, and the growth of the vaterite particles occurs through a dissolution-reprecipitation mechanism of nanoparticles. While these studies did not measure the reactive surface area, it is possible that such pitting and the presence of nanoparticles increases the reactive surface area and enhances dissolution under flowing conditions.

[35] Our model allows us to simulate and mechanistically evaluate the precipitation and dissolution of CaCO₃ that was observed in a micromodel pore network. The overall rate and amount of precipitation are reasonably captured and mechanistically explained, but not the enhanced dissolution that occurs due to pore blockage and reduced mixing. We note that the reactive surface area may increase during dissolution due to nanometer scale features such as pitting and the presence of nanoparticles. However, direct evidence of this process in our system is lacking. Further work is needed to mechanistically evaluate the reason for enhanced dissolution in the micromodel experiment. This has implications for geological carbon sequestration, because the extent of precipitation at all times during CO₂ injection may affect pore blockage, the efficiency of CO₂ injection, and the ability to mix and sequester CO₂ in the geologic formation.

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